GenAi Training Day 6

Multi-Modal LLM

Overview

- LLMs are no longer text-only systems
- Modern models can see, hear, and speak
- Enable richer, more natural human–AI interaction

What Are Multi-Modal LLMs?

- Process and generate multiple data types
- Modalities include:
  - Text
  - Images
  - Audio
  - Speech
- Go beyond chatbots to intelligent interfaces

Why Multi-Modality Matters

- Humans communicate using multiple senses
- Most real-world data is not text
- Multi-modal AI feels more natural and human
- Combining modalities improves understanding

Modalities Overview

- **Text:** reasoning, summarization, coding
- **Vision:** images, documents, charts, UI screenshots
- **Audio:** speech recognition, translation, tone
- **Speech:** voice-based interaction

Vision Capabilities

- Image understanding & object detection
- **OCR:** extract text from images and documents
- Image Q&A and document analysis
- Diagram, chart, and UI understanding

Audio Understanding

- **Speech-to-Text (STT**) for commands and transcription
- **Text-to-Speech (TTS)** for natural voice output
- Real-time transcription and translation
- Example tools: Whisper, GPT-4o Transcribe

Voice Assistant Flow

- User speaks
- **STT** converts audio to text
- **LLM** reasons over text
- **TTS** converts response to speech

Multi-Modal Examples

- **Gemini:** text, image, audio, video understanding
- **DALL·E:** text-to-image generation
- **Use cases:** design, accessibility, automation

Benefits & Challenges

- **Benefits**
  - Better context understanding
  - Improved accessibility
  - Visual and voice workflow automation
- **Challenges**
  - Higher cost and latency
  - Complex pipelines
  - Safety across modalities

Safety Considerations

- Image and audio content moderation
- Privacy and sensitive data handling
- Guardrails remain essential